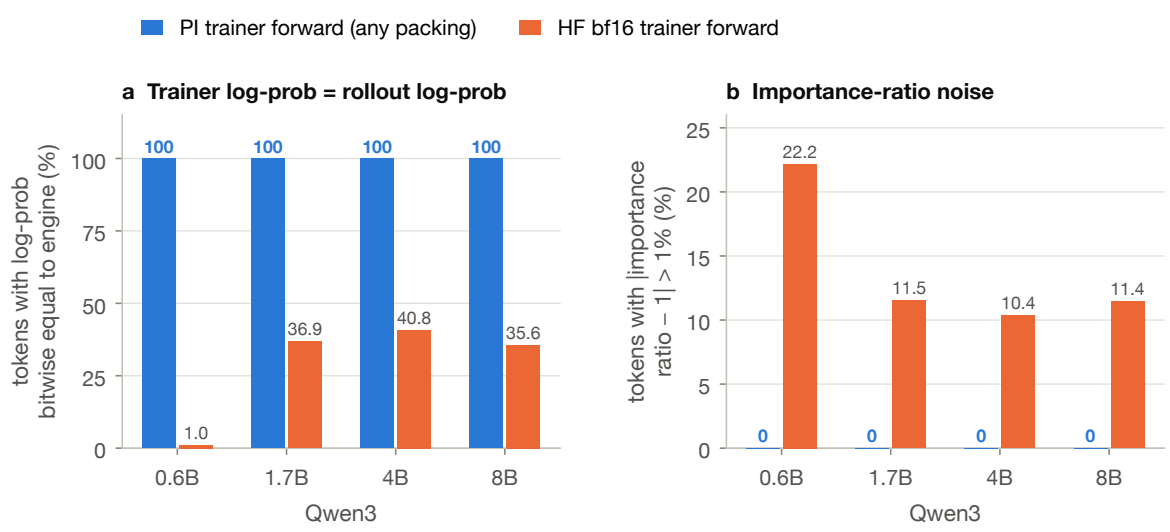
PI trainer forward (any packing)
HF bf16 trainer forward
a Trainer log-prob = rollout log-prob
tokens with log-prob bitwise equal to engine (%)
100
75
50
25
0
100
100
100
100
1.0
36.9
40.8
35.6
0.6B
1.7B
4B
8B
Qwen3
b Importance-ratio noise
tokens with |importance ratio − 1| > 1% (%)
25
20
15
10
5
0
22.2
11.5
10.4
11.4
0
0
0
0
0.6B
1.7B
4B
8B
Qwen3